

What is rooting?

Rooting a phone is Android-speak for gaining administrator-level privileges to the phone. Typically, users don't have access to a phone's higher level files and folders and can only deal with files and directories stored on the phone's external memory or SD card. The system critical files, processes and applications like the dialer, messaging and camera apps reside in the phone's internal memory. Users can't uninstall these apps, nor can they directly make changes to the filesystem. This is where rooting comes in. By rooting one's phone, the user is granted elevated privileges and can make modifications to any file or applications located anywhere on the filesystem.

As a crude analogy, consider your desktop computer running Windows – Your 'Windows' folder (typically located under C:\Windows) contains system files which are critical to the system. Not having root privileges on an Android device is analogous to not being able to edit the contents of this Windows folder. Rooting one's device would be similar to 'unlocking' the contents of the Windows folder and being granted the ability to add, remove or edit files at your will. While this may not seem like a big deal (As most of us don't interact directly with our Windows folders) there are several apps that can take advantage of this ability, resulting in greater functionality for your device. (This analogy has several inaccuracies, least of all that Windows uses a very different filesystem structure than Android and other UNIX-based systems, which use a top-down or tree-like hierarchy.)

The term 'root' is a feature of UNIX and UNIX-based systems which use a tree-like filesystem structure. The top level directory is designated with a '/' and is called root. Having access to files at this level of the hierarchy is having 'root' access to the system, and a user having root access is a root user (or superuser). Since Android uses a Linux kernel, and Linux itself is derived from UNIX, the terminology carries on.

How does the rooting process work?

In order to understand how rooting occurs, we need some insight into the Android system.

Most devices, by default have their /system partition (found in the device's internal memory) set to read-only, which means apps cannot modify this sacred folder. Mounting the /system partition with read-write access requires root privileges. In an Android system, only a privileged app (i.e. An app running as root) can invoke or start another privileged app. Ordinary apps that are run as user cannot start apps that need privileged